Kernix 0.12.0 / Performance brief

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

46 workloads

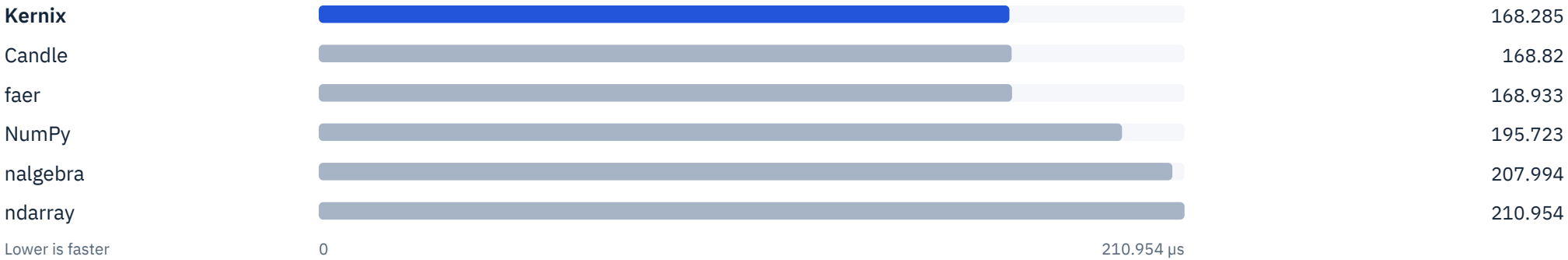
152 observations

Source precision retained

Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

Kernix 0.12.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication · transposed view

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · borrowed transpose · allocated output · independent operands



Matrix multiplication · transposed view

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · borrowed transpose · allocated output · independent operands



Kernix 0.12.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



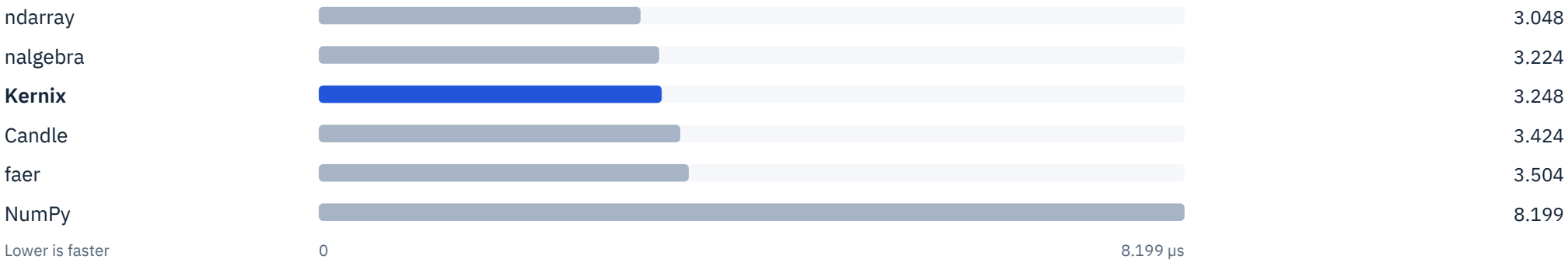
Kernix 0.12.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



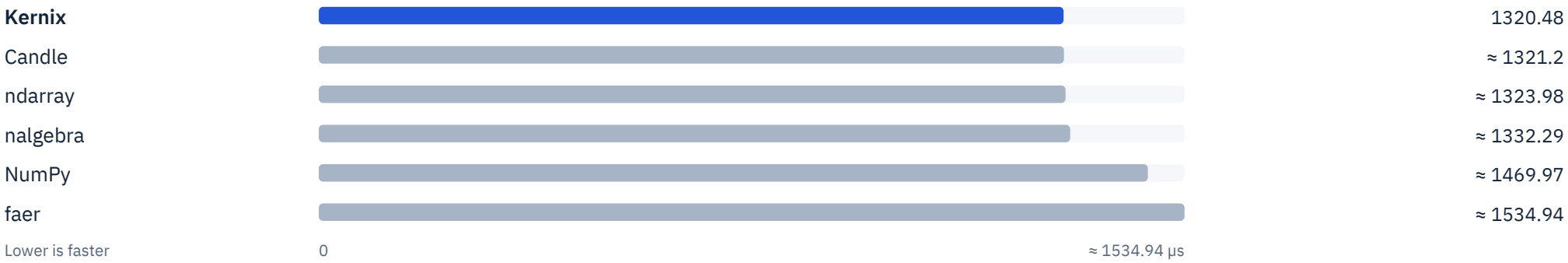
Kernix 0.12.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

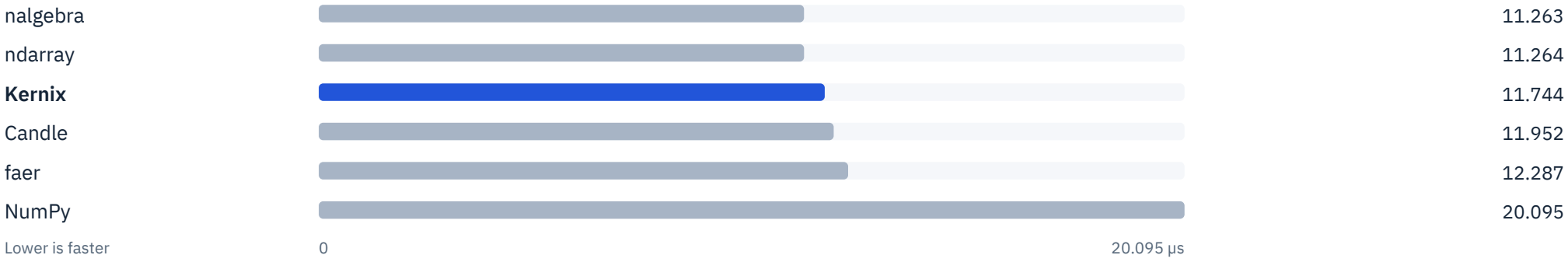
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



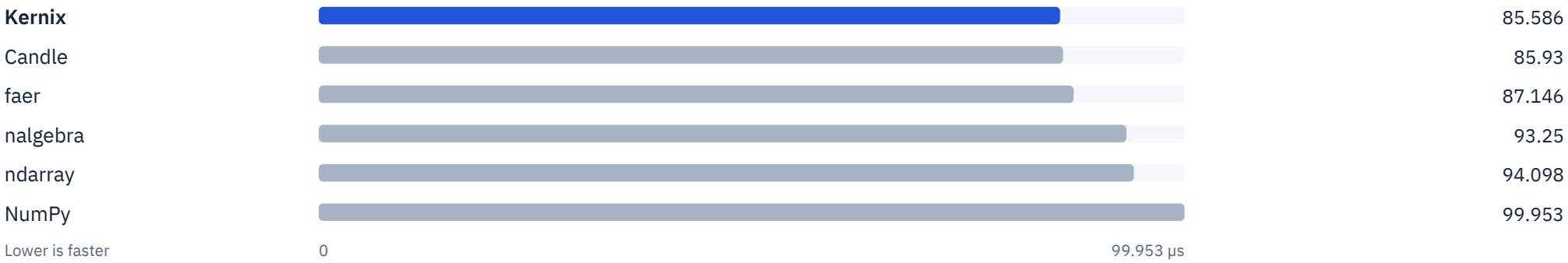
Kernix 0.12.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

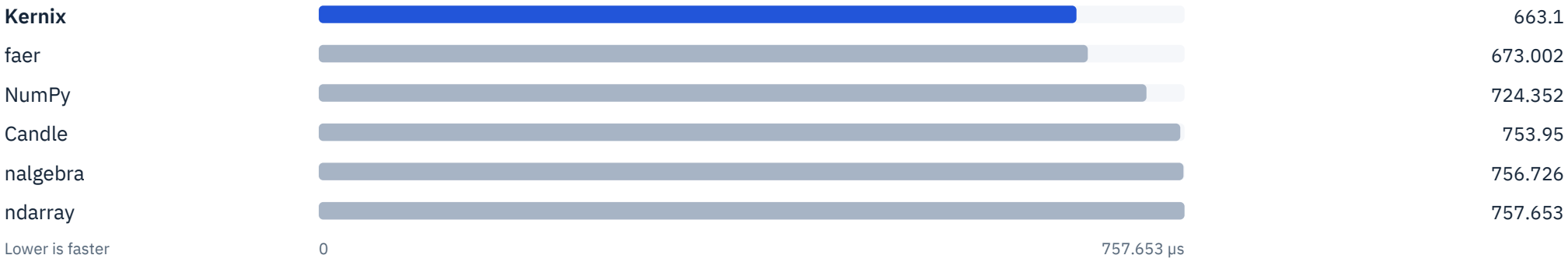
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



Kernix 0.12.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

Per-call median

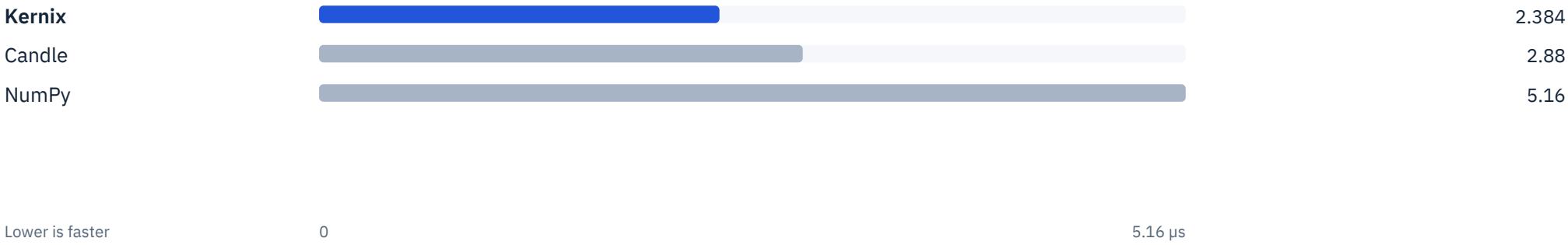
4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f32 · 1 worker · allocated output · broadcast batches



Kernix 0.12.0 / Explicit parallel execution

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.12.0 / Explicit parallel execution

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



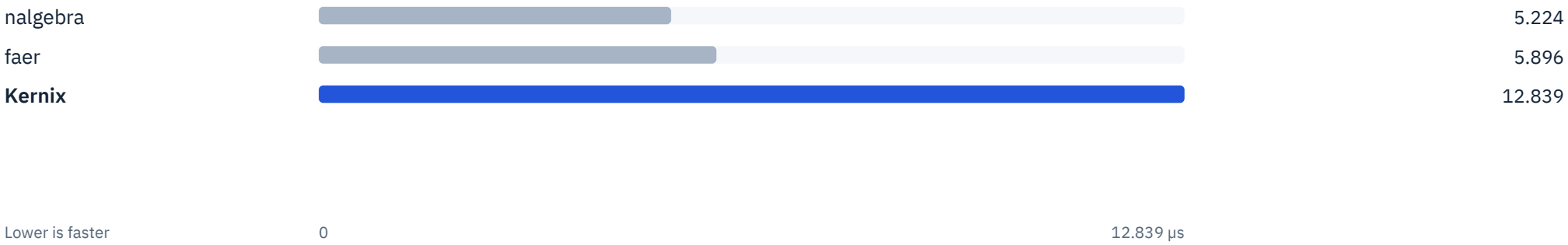
Kernix 0.12.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky solve · reused factor

Per-call median

128 × 128, 1 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



Cholesky solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



Kernix 0.12.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky solve · reused factor

Per-call median

512 × 512, 128 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



LU solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



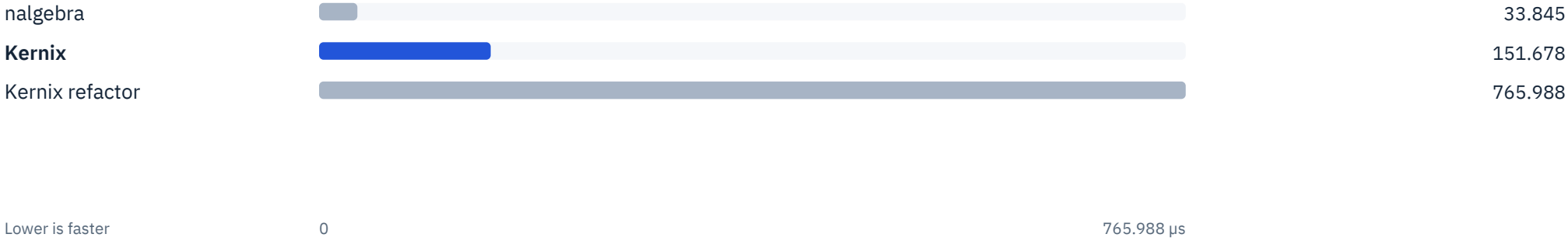
Kernix 0.12.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky rank-one update

256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix

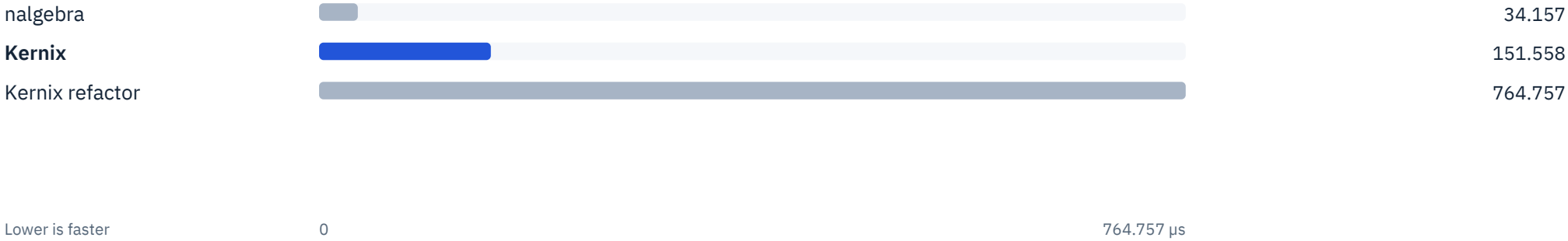
Per-call median



Cholesky rank-one downdate

256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix

Per-call median



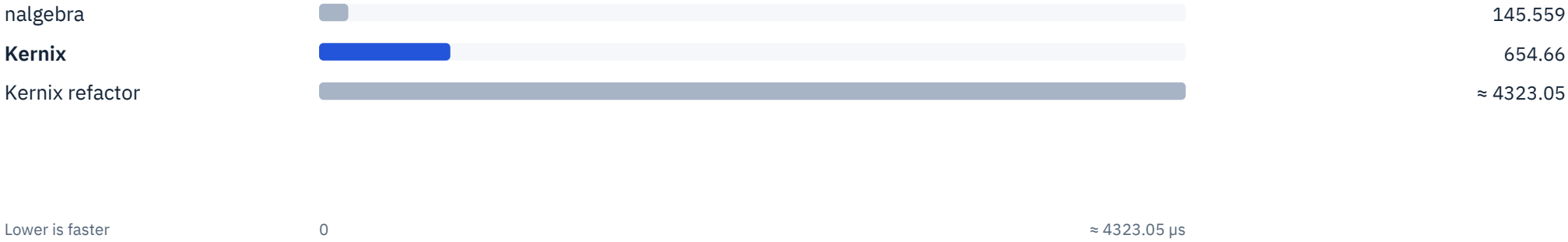
Kernix 0.12.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky rank-one update

512 × 512 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix

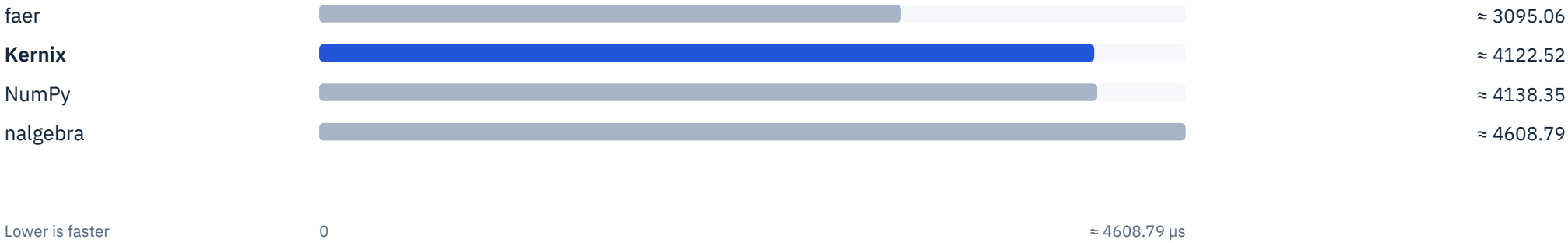
Per-call median



Thin QR

512 × 128 · f64 · 1 worker · allocated output

Per-call median



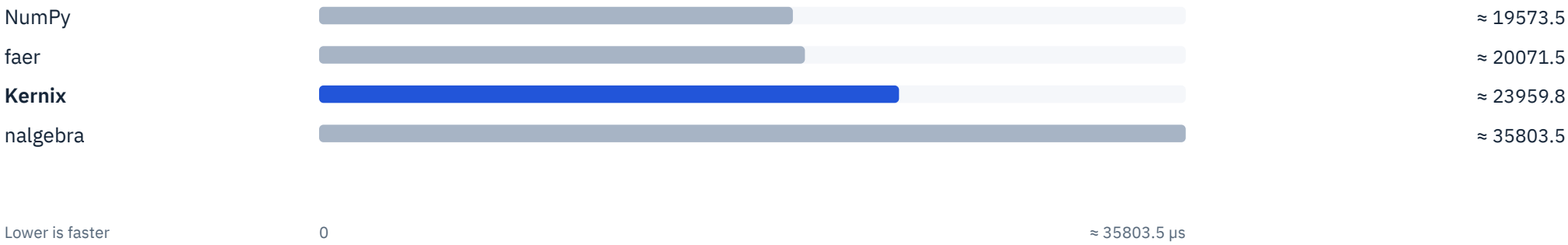
Kernix 0.12.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

1024 × 256 · f64 · 1 worker · allocated output

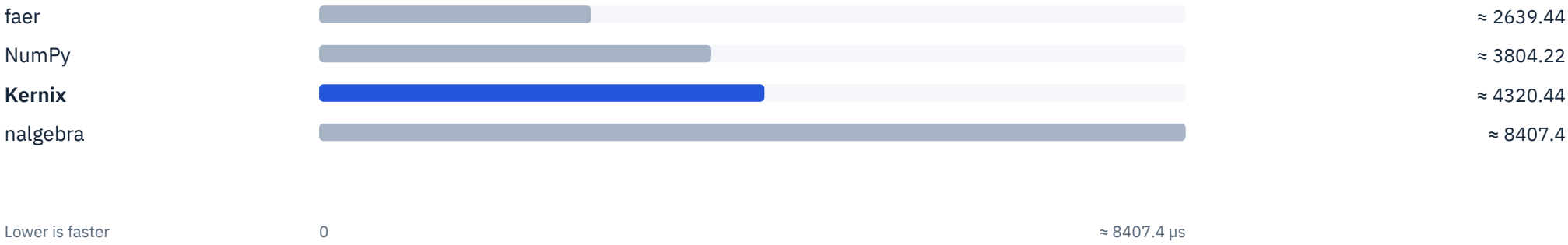
Per-call median



Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output

Per-call median



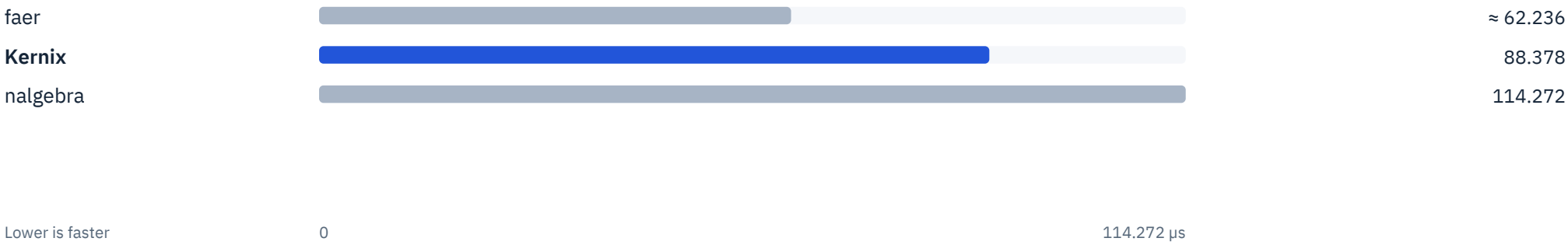
Kernix 0.12.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output

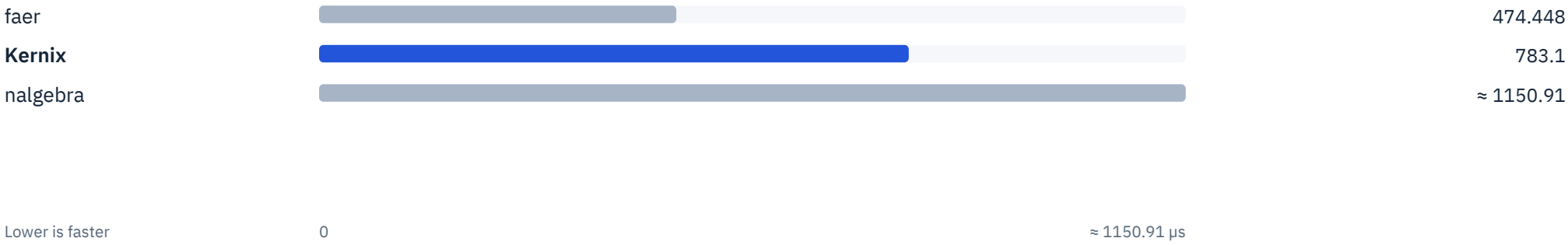
Per-call median



Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



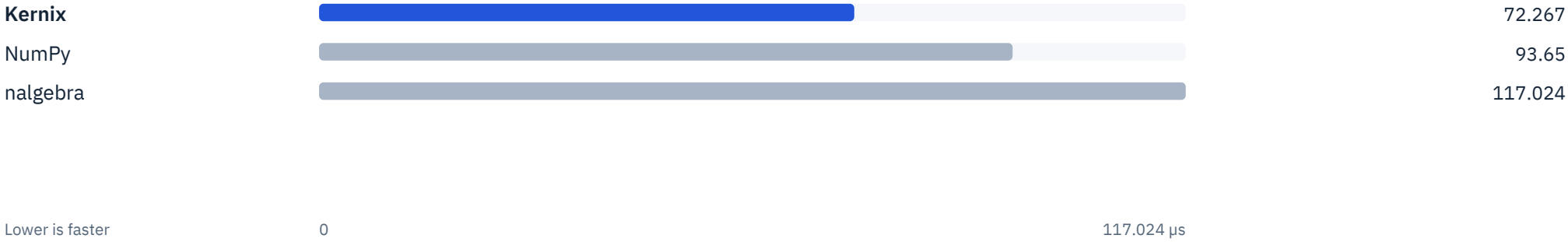
Kernix 0.12.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

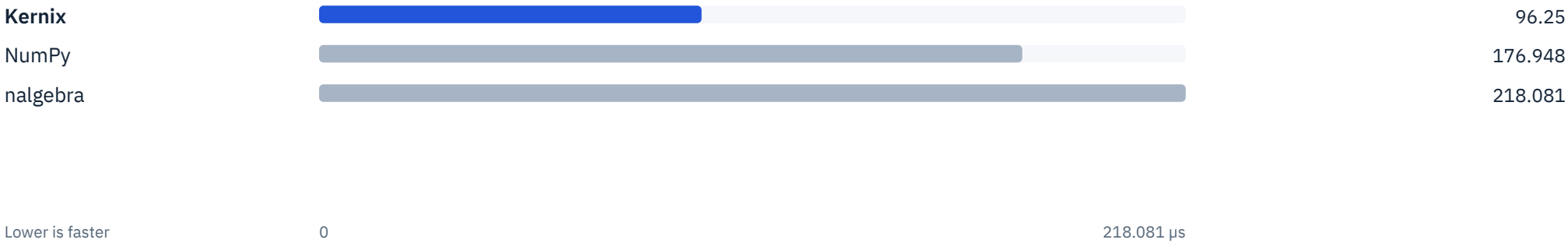
64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Kernix 0.12.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

96 × 32 · f64 · 1 worker · allocated output

Per-call median



Thin QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.12.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output

Per-call median



Cholesky factorization

128 × 128 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.12.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output



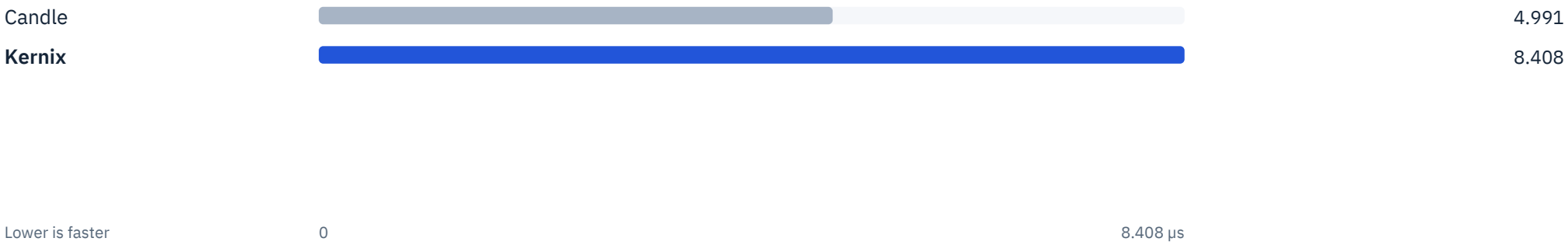
Kernix 0.12.0 / Training & autodiff

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Kernix 0.12.0 / Training & autodiff

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.12.0 / Training & autodiff

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented



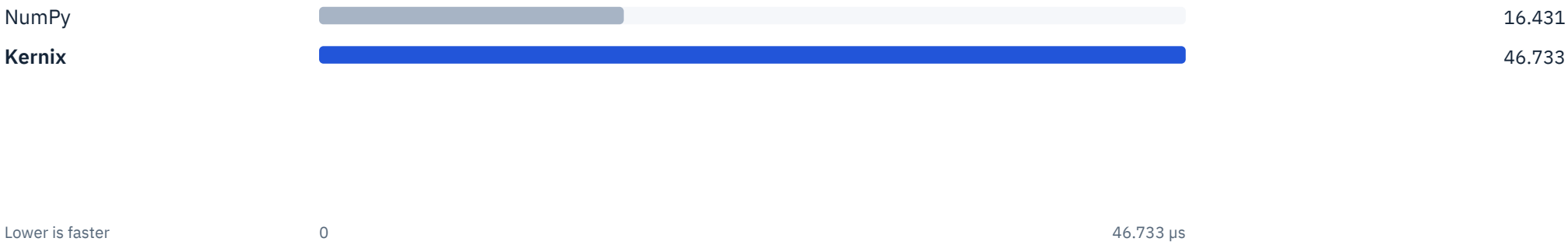
Kernix 0.12.0 / Tensor contractions

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.12.0 / Statistics & activations

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output

Per-call median

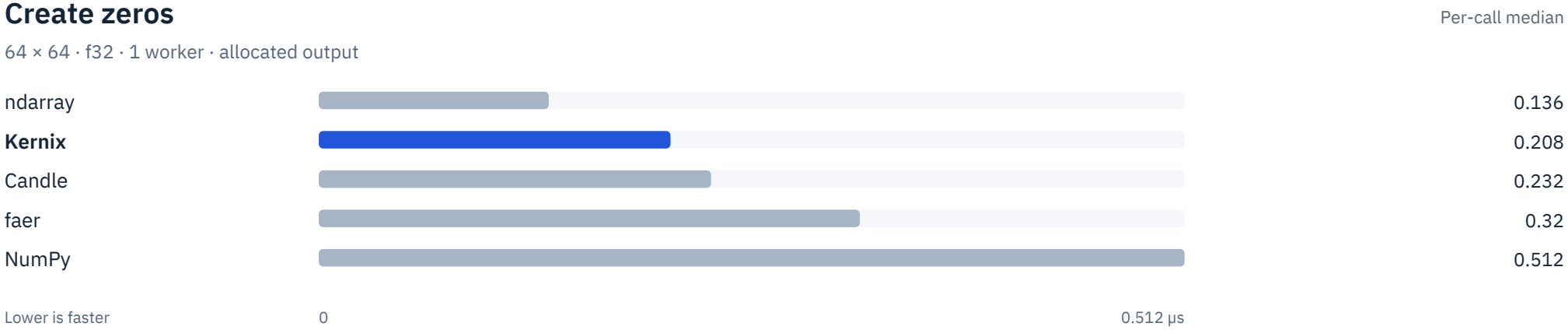


Kernix 0.12.0 / Tensor construction

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

64 × 64 · f32 · 1 worker · allocated output



Create ones

64 × 64 · f32 · 1 worker · allocated output



Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Cholesky solve · reused factor

128 × 128, 1 RHS · f64

Observed time increase



CURRENT / BASELINE	95% INTERVAL
1.0234986945169713	1.0037549870922318 – 1.028179190751445

Cholesky solve · reused factor

256 × 256, 64 RHS · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.33792112617530035	0.3375684915229404 – 0.3384791160852235

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

LU solve · reused factor

256 × 256, 64 RHS · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.47913481363078275	0.47624069058659213 – 0.48246535292720216

Thin QR

512 × 128 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.8726855191223079	0.8695658207967547 – 0.8755869170594145

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Thin QR

1024 × 256 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.6789324478702599	0.6782241501332267 – 0.6818595664936169

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9997738334754727	0.9953238460306846 – 1.0194679976178997

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9970730657484128	0.9917739551423491 – 1.0087548004946951

Thin QR

256 × 64 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9995848863933859	0.9948733492097857 – 1.0002446010470083

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

32 × 32 · 32 × 32 · f64

Timing direction unclear



CURRENT / BASELINE	1	95% INTERVAL	0.995085995085995 – 1.0003087372645878

Matrix multiplication

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	**0.9988561419813765**	95% INTERVAL	0.9961922526300783 – 1.0001961530231342

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

128 × 128 · 128 × 128 · f32

Observed time increase



CURRENT / BASELINE	95% INTERVAL
1.002169477214626	1.0002808134228816 – 1.0036371291239314

Matrix multiplication

256 × 256 · 256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0001756182785322	0.9991833617518403 – 1.0009622780930691

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

256 × 256 · 256 × 256 · f32

Observed time reduction



CURRENT / BASELINE	**0.9983208921448529**	95% INTERVAL	0.9957323002012497 – 0.9991901500488781

Record + backward

sum(A · W), 32 × 16 · 16 × 8 · f64

Timing direction unclear



CURRENT / BASELINE	**0.9954545454545455**	95% INTERVAL	0.9889674681753889 – 1.0002854695974879

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Independent matrix operands

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.998486398278278	0.9957582184517497 – 1.003750007416945

Independent matrix operands

128 × 128 · 128 × 128 · f32

Observed time increase



CURRENT / BASELINE	95% INTERVAL
1.0026257439607889	1.000093394662495 – 1.0043065616552176

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64

Observed time increase



CURRENT / BASELINE	95% INTERVAL
1.0041753653444678	1.0020876826722338 – 1.0062761506276152

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9995819397993311	0.9966555183946488 – 1

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

128 × 128 · f64

Observed time increase



CURRENT / BASELINE	95% INTERVAL
1.0067263843058276	1.000258549180004 – 1.0133176513163846

Cholesky factorization

256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9993030841457505	0.9967016127769105 – 1.0029723941355255

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

512 × 512 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.9962459765630284	0.994448678774562 – 0.999483314139962

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0091958049613556	0.9895708760248662 – 1.0212185378419087

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0036071534217699	0.997541024812381 – 1.006617502894442

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0003274059219547	0.9985314514155177 – 1.0022661848471857

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f64	212186	212.186
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f64	197098	197.098
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f32	92713	92.713
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f32	100370	100.37
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	168484	168.484
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	195419	195.419
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	86258	86.258
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	99857	99.857
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	3424	3.424
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	8199	8.199

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	3048	3.048
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	3504	3.504
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	3224	3.224
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	3248	3.248
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	22615	22.615
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	22455	22.455
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	22590	22.59
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	21815	21.815
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	31046	31.046
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	22167	22.167

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	168285	168.285
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	207994	207.994
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	168933	168.933
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	168820	168.82
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	210954	210.954
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	195723	195.723
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	1534937	1534.937
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	1332286	1332.286
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	1469973	1469.973
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	1323982	1323.982

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	1321199	1321.199
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	1320480	1320.48
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	11744	11.744
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	11952	11.952
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	20095	20.095
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	12287	12.287
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	11263	11.263
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	11264	11.264
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	85586	85.586
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	94098	94.098

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	87146	87.146
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	93250	93.25
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	99953	99.953
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	85930	85.93
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	756726	756.726
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	663100	663.1
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	673002	673.002
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	724352	724.352
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	757653	757.653
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	753950	753.95

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	12991	12.991
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	3855.0000000000005	3.8550000000000005
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	4432	4.432
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f32	2384	2.384
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f32	5160	5.16
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f32	2880	2.88
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	16798	16.798
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	92914	92.914
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	698185	698.185
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	49044	49.044

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky solve · reused factor / faer 128 × 128, 1 RHS · f64	5896	5.896
Cholesky solve · reused factor / nalgebra 128 × 128, 1 RHS · f64	5224	5.224
Cholesky solve · reused factor / Kernix 128 × 128, 1 RHS · f64	12839	12.839
Cholesky solve · reused factor / Kernix 256 × 256, 64 RHS · f64	1718605	1718.605
Cholesky solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	1276226	1276.226
Cholesky solve · reused factor / faer 256 × 256, 64 RHS · f64	448842	448.842
Cholesky solve · reused factor / Kernix 512 × 512, 128 RHS · f64	13442704	13442.704
Cholesky solve · reused factor / faer 512 × 512, 128 RHS · f64	3181250	3181.25
Cholesky solve · reused factor / nalgebra 512 × 512, 128 RHS · f64	12652581	12652.581
LU solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	1228773	1228.773

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
LU solve · reused factor / Kernix 256 × 256, 64 RHS · f64	1652017	1652.017
LU solve · reused factor / faer 256 × 256, 64 RHS · f64	454778	454.778
Cholesky rank-one update / nalgebra 256 × 256 · f64	33845	33.845
Cholesky rank-one update / Kernix refactor 256 × 256 · f64	765988	765.988
Cholesky rank-one update / Kernix 256 × 256 · f64	151678	151.678
Cholesky rank-one downdate / nalgebra 256 × 256 · f64	34157	34.157
Cholesky rank-one downdate / Kernix refactor 256 × 256 · f64	764757	764.757
Cholesky rank-one downdate / Kernix 256 × 256 · f64	151558	151.558
Cholesky rank-one update / Kernix refactor 512 × 512 · f64	4323053	4323.053
Cholesky rank-one update / nalgebra 512 × 512 · f64	145559	145.559

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky rank-one update / Kernix 512 × 512 · f64	654660	654.66
Thin QR / Kernix 512 × 128 · f64	4122523.0000000005	4122.5230000000005
Thin QR / nalgebra 512 × 128 · f64	4608794	4608.794
Thin QR / NumPy 512 × 128 · f64	4138346	4138.346
Thin QR / faer 512 × 128 · f64	3095064	3095.064
Thin QR / faer 1024 × 256 · f64	20071459	20071.459
Thin QR / NumPy 1024 × 256 · f64	19573491	19573.491
Thin QR / nalgebra 1024 × 256 · f64	35803543	35803.543
Thin QR / Kernix 1024 × 256 · f64	23959756	23959.756
Cholesky factorization / NumPy 512 × 512 · f64	3804216	3804.216

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / nalgebra 512 × 512 · f64	8407403	8407.403
Cholesky factorization / faer 512 × 512 · f64	2639439	2639.439
Cholesky factorization / Kernix 512 × 512 · f64	4320437	4320.437
Column-pivoted QR / faer 96 × 32 · f64	62236.00000000001	62.23600000000001
Column-pivoted QR / Kernix 96 × 32 · f64	88378	88.378
Column-pivoted QR / nalgebra 96 × 32 · f64	114272	114.272
Column-pivoted QR / nalgebra 256 × 64 · f64	1150907	1150.907
Column-pivoted QR / Kernix 256 × 64 · f64	783100	783.1
Column-pivoted QR / faer 256 × 64 · f64	474448	474.448
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	72267	72.267

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	117024	117.024
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	93650	93.65
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	218081	218.081
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	96250	96.25
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	176948	176.948
Thin QR / nalgebra 96 × 32 · f64	61308	61.308
Thin QR / NumPy 96 × 32 · f64	66027	66.027
Thin QR / faer 96 × 32 · f64	65451.99999999999	65.45199999999999
Thin QR / Kernix 96 × 32 · f64	73411	73.411
Thin QR / NumPy 256 × 64 · f64	565426	565.426

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / nalgebra 256 × 64 · f64	586225	586.225
Thin QR / faer 256 × 64 · f64	451250	451.25
Thin QR / Kernix 256 × 64 · f64	693442	693.442
Cholesky factorization / faer 64 × 64 · f64	13247	13.247
Cholesky factorization / Kernix 64 × 64 · f64	27374	27.374
Cholesky factorization / nalgebra 64 × 64 · f64	20334	20.334
Cholesky factorization / NumPy 64 × 64 · f64	17983	17.983
Cholesky factorization / faer 128 × 128 · f64	65236	65.236
Cholesky factorization / NumPy 128 × 128 · f64	82282	82.282
Cholesky factorization / Kernix 128 × 128 · f64	279326	279.326

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / nalgebra 128 × 128 · f64	128896.00000000001	128.89600000000001
Cholesky factorization / Kernix 256 × 256 · f64	763797	763.797
Cholesky factorization / nalgebra 256 × 256 · f64	1029890.0000000001	1029.8900000000001
Cholesky factorization / NumPy 256 × 256 · f64	748773	748.773
Cholesky factorization / faer 256 × 256 · f64	403316	403.316
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	4991	4.991
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	8408	8.408
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	7183	7.183
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	3519	3.519
Model inference / Candle batched normalized/sliced MLP · f64	5280	5.28

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model inference / Kernix batched normalized/sliced MLP · f64	3048	3.048
Model recording + forward / Kernix batched normalized/sliced MLP · f64	3848.0000000000005	3.8480000000000005
Model recording + forward / Candle batched normalized/sliced MLP · f64	7607	7.607
Model backward / Kernix prepared graph → fresh gradients · f64	4183	4.183
Model backward / Candle prepared graph → fresh gradients · f64	21199	21.199
Parameter update / Kernix transactional SGD, fixed gradients · f64	744	0.744
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	46733	46.733
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	16431	16.431
Row softmax / Kernix 256 × 64 · f64	113008	113.008
Row softmax / ndarray 256 × 64 · f64	97465	97.465

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Row softmax / NumPy 256 × 64 · f64	134095	134.095
Row softmax / Candle 256 × 64 · f64	110384	110.384
Create zeros / NumPy 64 × 64 · f32	512	0.512
Create zeros / ndarray 64 × 64 · f32	136	0.136
Create zeros / Kernix 64 × 64 · f32	208	0.208
Create zeros / faer 64 × 64 · f32	320	0.32
Create zeros / Candle 64 × 64 · f32	232	0.232
Create ones / faer 64 × 64 · f32	239.99999999999997	0.23999999999999997
Create ones / Candle 64 × 64 · f32	344	0.344
Create ones / NumPy 64 × 64 · f32	1520	1.52

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create ones / ndarray 64 × 64 · f32	208	0.208
Create ones / Kernix 64 × 64 · f32	256	0.256